Unit 4 — Relations

1. Definition of a Relation

A relation is a mathematical way of describing a connection or association between elements of two sets.

In simple words: A relation tells us which elements of one set are "related to" which elements of another set.

Formal Definition: Let A and B be two non-empty sets. A relation R from set A to set B is defined as any subset of the Cartesian product A × B.

That is: $R \subseteq A \times B$

If $(a, b) \in R$, we say "a is related to b" and write it as a R b. If $(a, b) \notin R$, then a is not related to b.

Example:

Let A = {1, 2, 3} and B = {4, 5}

Cartesian product: A × B = {(1,4), (1,5), (2,4), (2,5), (3,4), (3,5)}

One possible relation R from A to B: R = {(1,4), (2,5), (3,4)}

This means: 1 is related to 4, 2 is related to 5, and 3 is related to 4.

Important Note: Every relation is a subset of A × B, but not every subset needs to follow any specific rule. A relation can be any collection of ordered pairs chosen from A × B.

2. Binary Relation

Definition: A binary relation on a set A is a relation from A to A itself.

That is: $R \subseteq A \times A$

It is called "binary" because it connects two elements from the same set.

Notation: If $(a, b) \in R$, we write: a R b (read as "a is related to b")

Example:

Let A = {1, 2, 3}

A × A = {(1,1), (1,2), (1,3), (2,1), (2,2), (2,3), (3,1), (3,2), (3,3)}

Define R = {(1,1), (2,2), (3,3)}

Here, each element is related to itself. This is a binary relation on A.

Real-Life Meaning: Think of "is a friend of" in a group of people. This is a binary relation because it connects people from the same set (same group) to each other.

Key Point: A binary relation R on set A with n elements can have anywhere from 0 to n^2 ordered pairs, since $|A \times A| = n^2$.

3. Representation of Relations

A relation can be represented in four different ways. All four are equivalent — they just show the same information in different forms.

3.1 Set Representation (Roster Form)

Simply list all the ordered pairs that belong to the relation.

Example:

Let A = {1, 2, 3} R = "less than" relation on A R = {(1,2), (1,3), (2,3)}

3.2 Matrix Representation (Boolean Matrix / Relation Matrix)

If a relation R is defined on set A = $\{a_1, a_2, \ldots, a_n\}$, we create an n × n matrix M where:

M[i][j] = 1, if $(a_i, a_j) \in R$ M[i][j] = 0, if $(a_i, a_j) \notin R$

This matrix is called the Boolean matrix or bit matrix of the relation.

Example:

Let A = {1, 2, 3} R = {(1,1), (1,2), (2,3), (3,1)}

Matrix M:

	1	**2**	**3**
1	1	1	0
2	0	0	1
3	1	0	0

Reading the matrix: Row 1, Column 2 = 1 → (1,2) ∈ R ✔ Row 2, Column 1 = 0 → (2,1) ∉ R ✔

Properties visible from matrix:

- If the main diagonal has all 1s → relation is reflexive
- If the matrix equals its transpose → relation is symmetric

3.3 Digraph Representation (Directed Graph)

Each element of A is drawn as a vertex (circle/node). For each ordered pair (a, b) ∈ R, draw a directed arrow from vertex a to vertex b.

If (a, a) ∈ R (element related to itself), draw a loop at vertex a.

Example:

Let A = {1, 2, 3} R = {(1,2), (2,3), (3,1)}

Digraph: 1 → 2 → 3 → 1 (forms a cycle)

Reading the digraph: An arrow from a to b means (a, b) ∈ R.

3.4 Arrow Diagram (for relation from A to B)

When R is a relation from set A to a different set B, draw two separate ovals — one for A and one for B. Draw arrows from elements of A to elements of B based on which pairs belong to R.

Example:

A = {1, 2, 3}, B = {a, b} R = {(1,a), (2,b), (3,a)}

Arrow from 1 → a, 2 → b, 3 → a

4. Domain and Range of a Relation

Domain: The domain of a relation R from A to B is the set of all first elements (left-side elements) of the ordered pairs in R.

Domain(R) = {a ∈ A | (a, b) ∈ R for some b ∈ B}

Range: The range of a relation R from A to B is the set of all second elements (right-side elements) of the ordered pairs in R.

Range(R) = {b ∈ B | (a, b) ∈ R for some a ∈ A}

Example:

Let R = {(1, a), (2, b), (2, c), (3, a)}

Domain(R) = {1, 2, 3} Range(R) = {a, b, c}

Important Observation: Domain(R) ⊆ A and Range(R) ⊆ B. The domain may not include every element of A if some elements of A are not related to anything. Similarly, the range may not include every element of B.

Example where domain ≠ A:

A = {1, 2, 3, 4}, B = {x, y} R = {(1, x), (2, y)}

Domain(R) = {1, 2} ← 3 and 4 are not in domain Range(R) = {x, y}

5. Types of Relations

5.1 Universal Relation

Definition: A universal relation on a set A is a relation in which every element of A is related to every element of A. It is simply the full Cartesian product A × A.

R = A × A

Every possible ordered pair is included.

Example:

A = {1, 2} A × A = {(1,1), (1,2), (2,1), (2,2)} Universal Relation R = {(1,1), (1,2), (2,1), (2,2)}

Key Point: The universal relation is the largest possible relation on A. It includes all n^2 ordered pairs where n = |A|.

Real-Life Analogy: In a group of people, "is a human being like" — everyone is related to everyone.

5.2 Void Relation (Empty Relation)

Definition: A void relation (also called an empty relation) on a set A is a relation that contains no ordered pairs at all.

R = ∅ (empty set)

No element of A is related to any element.

Example:

A = {1, 2, 3} R = ∅

No pair exists in R, so no element is related to anything.

Key Point: The void relation is the smallest possible relation on A. It is a valid subset of A × A (the empty subset).

Real-Life Analogy: "Is a sibling of" in a group of only children — nobody is related to anyone.

5.3 Identity Relation

Definition: The identity relation on set A is a relation where every element is related only to itself.

I = {(a, a) | a ∈ A}

Example:

A = {1, 2, 3} Identity Relation I = {(1,1), (2,2), (3,3)}

In matrix form: The identity relation gives the identity matrix (all 1s on diagonal, 0s elsewhere).

Comparison Table:

Type	Ordered Pairs	Size
Void Relation	None	0 pairs
Identity Relation	Only (a, a) pairs	n pairs
Universal Relation	All possible pairs	n^2 pairs

6. Operations on Relations

Just like sets support operations (union, intersection, complement), relations also support these same operations since relations are themselves sets of ordered pairs.

Let R and S be two relations from set A to set B.

6.1 Union of Relations

Definition: The union of two relations R and S, denoted R ∪ S, is the set of all ordered pairs that belong to R or S or both.

$R \cup S = \{(a, b) \mid (a, b) \in R \text{ or } (a, b) \in S\}$

Example:

A = {1, 2, 3} R = {(1,2), (2,3)} S = {(2,3), (3,1)}

R ∪ S = {(1,2), (2,3), (3,1)}

(2,3) appeared in both but is written only once in the union.

Matrix Method: The matrix of R ∪ S is obtained by applying the OR operation element-by-element on the matrices of R and S.

M(R ∪ S) = M(R) OR M(S)

Where: 0 OR 0 = 0, and anything with 1 gives 1.

6.2 Intersection of Relations

Definition: The intersection of two relations R and S, denoted R ∩ S, is the set of all ordered pairs that belong to both R and S simultaneously.

$R \cap S = \{(a, b) \mid (a, b) \in R \text{ and } (a, b) \in S\}$

Example:

R = {(1,2), (2,3), (3,1)} S = {(2,3), (3,1), (1,3)}

R ∩ S = {(2,3), (3,1)}

Only pairs that appear in BOTH relations are included.

Matrix Method: The matrix of R ∩ S is obtained by applying the AND operation element-by-element.

M(R ∩ S) = M(R) AND M(S)

Where: 1 AND 1 = 1, and anything with 0 gives 0.

6.3 Complement of a Relation

Definition: The complement of a relation R on set A, denoted R' or $\bar{R}$ or ~R, is the set of all ordered pairs from A × A that are NOT in R.

$R' = (A \times A) - R = \{(a, b) \mid (a, b) \notin R\}$

Example:

A = {1, 2} A × A = {(1,1), (1,2), (2,1), (2,2)} R = {(1,2), (2,1)}

R' = {(1,1), (2,2)}

Matrix Method: The matrix of R' is obtained by flipping every bit — change all 0s to 1s and all 1s to 0s (NOT operation).

M(R') = NOT M(R)

Observation: R ∪ R' = A × A (Universal Relation) R ∩ R' = ∅ (Void Relation)

Summary Table — Operations on Relations

Operation	Symbol	Rule	Matrix Operation
Union	R ∪ S	Pair in R or S or both	M(R) OR M(S)
Intersection	R ∩ S	Pair in both R and S	M(R) AND M(S)
Complement	R'	Pair not in R	NOT M(R)

7. Properties of Binary Relations

These are the most important exam topics in this unit. A binary relation R on a set A can have the following key properties. Understanding these properties helps classify relations and build more advanced structures.

7.1 Reflexive Relation

Definition: A relation R on set A is called reflexive if every element of A is related to itself.

Formally: For all a ∈ A, (a, a) ∈ R

Every element must appear in a pair with itself.

Example 1 — Reflexive:

A = {1, 2, 3} R = {(1,1), (2,2), (3,3), (1,2)}

Check: (1,1) ∈ R ✔, (2,2) ∈ R ✔, (3,3) ∈ R ✔ All elements are self-related → R is reflexive.

Example 2 — Not Reflexive:

A = {1, 2, 3} R = {(1,1), (2,2)}

Check: (3,3) ∉ R ✗ Element 3 is not self-related → R is NOT reflexive.

Matrix Condition: A relation is reflexive if and only if ALL entries on the main diagonal of its matrix are 1.

Real-Life Example: "Is equal to" (=) — every number equals itself. So a = a for all a → reflexive. "Is greater than" (>) — 1 > 1 is false → NOT reflexive.

7.2 Irreflexive Relation

Definition: A relation R on set A is called irreflexive if NO element of A is related to itself.

Formally: For all a ∈ A, (a, a) ∉ R

Example:

A = {1, 2, 3} R = {(1,2), (2,3), (3,1)}

Check: (1,1) ∉ R ✔, (2,2) ∉ R ✔, (3,3) ∉ R ✔ → Irreflexive.

Matrix Condition: All diagonal entries are 0.

Note: A relation can be neither reflexive nor irreflexive if some (but not all) diagonal entries are 1.

7.3 Symmetric Relation

Definition: A relation R on set A is called symmetric if whenever (a, b) ∈ R, then (b, a) must also be in R.

Formally: For all a, b ∈ A, if (a, b) ∈ R → (b, a) ∈ R

Direction does not matter — if a is related to b, then b is automatically related to a.

Example 1 — Symmetric:

A = {1, 2, 3} R = {(1,2), (2,1), (2,3), (3,2)}

Check: (1,2) ∈ R and (2,1) ∈ R ✔ (2,3) ∈ R and (3,2) ∈ R ✔ → Symmetric.

Example 2 — Not Symmetric:

R = {(1,2), (2,3)}

Check: (1,2) ∈ R but (2,1) ∉ R ✗ → NOT symmetric.

Matrix Condition: A relation is symmetric if and only if its matrix equals its own transpose. $M(R) = M(R)^T$

That means: M[i][j] = M[j][i] for all i, j.

Real-Life Example: "Is a sibling of" — if A is a sibling of B, then B is a sibling of A → symmetric. "Is a parent of" — if A is a parent of B, B is NOT a parent of A → NOT symmetric.

7.4 Asymmetric Relation

Definition: A relation R is called asymmetric if whenever (a, b) ∈ R, then (b, a) must NOT be in R.

Formally: For all a, b ∈ A, if (a, b) ∈ R → (b, a) ∉ R

Also, (a, a) can never be in an asymmetric relation (it must be irreflexive too).

Example:

R = {(1,2), (2,3), (1,3)}

(1,2) ∈ R and (2,1) ∉ R ✔ (2,3) ∈ R and (3,2) ∉ R ✔ → Asymmetric.

Real-Life Example: "Is a child of" — if A is a child of B, then B cannot be a child of A.

7.5 Anti-Symmetric Relation

Definition: A relation R on set A is called anti-symmetric if whenever (a, b) ∈ R and (b, a) ∈ R, then it must be that a = b.

Formally: For all a, b ∈ A, if (a, b) ∈ R and (b, a) ∈ R → a = b

In simpler words: Two different elements a and b cannot be related in both directions. If (a, b) and (b, a) are both in R, then a and b must actually be the same element.

Example 1 — Anti-Symmetric:

A = {1, 2, 3} R = {(1,1), (1,2), (2,3), (1,3)}

Check all pairs: (1,2) ∈ R, is (2,1) ∈ R? No ✔ (2,3) ∈ R, is (3,2) ∈ R? No ✔ (1,3) ∈ R, is (3,1) ∈ R? No ✔ → Anti-Symmetric.

Example 2 — Not Anti-Symmetric:

R = {(1,2), (2,1)}

(1,2) ∈ R and (2,1) ∈ R, but 1 ≠ 2 ✗ → NOT anti-symmetric.

Matrix Condition: If M[i][j] = 1 and i ≠ j, then M[j][i] must be 0. There should be no off-diagonal position where both M[i][j] = 1 and M[j][i] = 1.

Real-Life Example: "Is less than or equal to" (≤) — if a ≤ b and b ≤ a, then a = b → anti-symmetric. "Is a divisor of" — if a divides b and b divides a, then a = b → anti-symmetric.

Key Distinction — Symmetric vs Anti-Symmetric:

Property	**Condition**
Symmetric	(a,b) ∈ R → (b,a) ∈ R always
Anti-Symmetric	(a,b) ∈ R and (b,a) ∈ R → a = b
Asymmetric	(a,b) ∈ R → (b,a) ∉ R always

Note: A relation CAN be both symmetric and anti-symmetric at the same time. This happens only when R contains only pairs of the form (a, a) — i.e., only self-pairs.

7.6 Transitive Relation

Definition: A relation R on set A is called transitive if whenever (a, b) ∈ R and (b, c) ∈ R, then (a, c) must also be in R.

Formally: For all a, b, c ∈ A, if (a, b) ∈ R and (b, c) ∈ R → (a, c) ∈ R

Think of it as: "if A is connected to B, and B is connected to C, then A must be directly connected to C."

Example 1 — Transitive:

A = {1, 2, 3} R = {(1,2), (2,3), (1,3)}

Check: (1,2) ∈ R and (2,3) ∈ R → is (1,3) ∈ R? Yes ✔ → Transitive.

Example 2 — Not Transitive:

R = {(1,2), (2,3)}

Check: (1,2) ∈ R and (2,3) ∈ R → is (1,3) ∈ R? No ✗ → NOT transitive.

Example 3 — Transitive (Tricky case):

R = {(1,2)}

There is no pair (a, b) and (b, c) with b = 2 as a first element, so the condition is never triggered. → R is vacuously transitive.

Matrix Condition: R is transitive if M(R) × M(R) (Boolean matrix multiplication) has no entry that is 1 when the corresponding entry in M(R) is 0.

That is: if M^2 has a 1 at position (i, j), then M must also have a 1 at position (i, j).

Real-Life Example: "Is an ancestor of" — if A is an ancestor of B, and B is an ancestor of C, then A is an ancestor of C → transitive. "Is a friend of" — A is friends with B, B is friends with C, but A may NOT be friends with C → NOT necessarily transitive.

Summary Table — Properties of Binary Relations

Property	Condition	Matrix Clue	Real-Life Example
Reflexive	(a,a) ∈ R for all a	All diagonal = 1	"is equal to"
Irreflexive	(a,a) ∉ R for all a	All diagonal = 0	"is greater than"
Symmetric	(a,b) ∈ R → (b,a) ∈ R	$M = M^T$	"is a sibling of"
Asymmetric	(a,b) ∈ R → (b,a) ∉ R	No symmetric 1s	"is a child of"
Anti-Symmetric	(a,b),(b,a) ∈ R → a=b	No off-diagonal symmetric 1s	"≤" relation
Transitive	(a,b),(b,c) ∈ R → (a,c) ∈ R	M^2 entries ⊆ M entries	"is an ancestor of"

Quick Trick to Check Properties in Exams

Step 1 — Write the matrix of the relation. Step 2 — Check diagonal for reflexive / irreflexive. Step 3 — Compare matrix with its transpose for symmetric / anti-symmetric. Step 4 — Check all chains (a→b→c) and confirm (a,c) exists for transitive.

8. Partition and Covering of a Set

8.1 Covering of a Set

Definition: A collection of subsets $\{A_1, A_2, A_3, \ldots, A_n\}$ of a set A is called a covering of A if every element of A belongs to at least one of the subsets.

Formally: $A_1 \cup A_2 \cup \ldots \cup A_n = A$

The subsets in a covering may overlap with each other — shared elements are allowed.

Example:

$A = \{1, 2, 3, 4, 5\}$ $A_1 = \{1, 2, 3\}$ $A_2 = \{2, 3, 4\}$ $A_3 = \{4, 5\}$

$A_1 \cup A_2 \cup A_3 = \{1, 2, 3, 4, 5\} = A$ ✔

This is a valid covering. Note that 2, 3 appear in both A_1 and A_2 (overlap allowed).

8.2 Partition of a Set

Definition: A partition of a set A is a collection of non-empty subsets $\{A_1, A_2, \ldots, A_n\}$ of A such that:

Condition 1 — Coverage: Every element of A belongs to at least one subset. $A_1 \cup A_2 \cup \ldots \cup A_n = A$

Condition 2 — Disjointness: No two subsets share a common element. $A_i \cap A_j = \emptyset$ for $i \neq j$

Condition 3 — Non-emptiness: Each subset must be non-empty. $A_i \neq \emptyset$ for all i

Each subset A_i in the partition is called a block or cell of the partition.

Example 1 — Valid Partition:

$A = \{1, 2, 3, 4, 5, 6\}$ $A_1 = \{1, 2\}$, $A_2 = \{3, 4\}$, $A_3 = \{5, 6\}$

Coverage: $A_1 \cup A_2 \cup A_3 = \{1,2,3,4,5,6\} = A$ ✔ Disjoint: No shared elements ✔ Non-empty: All blocks have elements ✔ → Valid partition.

Example 2 — Not a Partition:

$A = \{1, 2, 3, 4\}$ $A_1 = \{1, 2, 3\}$, $A_2 = \{2, 3, 4\}$

Coverage: ✔ Disjoint: $A_1 \cap A_2 = \{2, 3\} \neq \emptyset$ ✗ → NOT a partition (overlap exists).

Key Difference — Covering vs Partition:

Property	Covering	Partition
Union = A	Required ✔	Required ✔
Overlap allowed	Yes ✔	No ✗
Empty subsets	Allowed	Not Allowed

Real-Life Example of Partition: Dividing a class of students into groups where each student belongs to exactly one group — no student in two groups, no student left out. That is a partition.

9. Equivalence Relation

9.1 Definition

A binary relation R on set A is called an equivalence relation if it satisfies all three of the following properties simultaneously:

1. Reflexive: (a, a) ∈ R for all a ∈ A
2. Symmetric: (a, b) ∈ R → (b, a) ∈ R
3. Transitive: (a, b) ∈ R and (b, c) ∈ R → (a, c) ∈ R

An equivalence relation groups elements that "behave the same way" under some criterion.

Example 1:

A = {1, 2, 3, 4} R = {(1,1), (2,2), (3,3), (4,4), (1,2), (2,1), (3,4), (4,3)}

Check: Reflexive: (1,1), (2,2), (3,3), (4,4) all present ✔ Symmetric: (1,2) ∈ R and (2,1) ∈ R ✔, (3,4) and (4,3) ✔ Transitive: (1,2) and (2,1) → need (1,1) ∈ R ✔. No chain produces a missing pair ✔ → R is an equivalence relation.

Example 2 — Real Life: "Congruence modulo n" on integers: a ≡ b (mod n) means a and b leave the same remainder when divided by n. This is always an equivalence relation.

9.2 Equivalence Classes

Definition: Let R be an equivalence relation on set A. For any element a ∈ A, the equivalence class of a, denoted [a], is the set of all elements in A that are related to a under R.

[a] = {b ∈ A | (a, b) ∈ R}

Key Properties of Equivalence Classes:

1. Every element of A belongs to exactly one equivalence class.
2. Two equivalence classes are either completely identical or completely disjoint — they never partially overlap.
3. The collection of all equivalence classes forms a partition of A.
4. [a] = [b] if and only if (a, b) ∈ R.

Example:

A = {1, 2, 3, 4, 5, 6} R = "same remainder when divided by 3"

Elements and their remainders: 1 mod 3 = 1 → Group: {1, 4} 2 mod 3 = 2 → Group: {2, 5} 0 mod 3 = 0 → Group: {3, 6}

Equivalence Classes: [1] = {1, 4} [2] = {2, 5} [3] = {3, 6}

Also note: [1] = [4], [2] = [5], [3] = [6] — same class, different representatives.

The three classes {1,4}, {2,5}, {3,6} form a partition of A. ✔

Connection to Partition: Every equivalence relation on A produces a partition of A into equivalence classes. Conversely, every partition of A defines an equivalence relation on A. This is the fundamental link between equivalence relations and partitions.

10. Compatibility Relation

10.1 Definition

A binary relation R on set A is called a compatibility relation if it satisfies only two properties:

1. Reflexive: $(a, a) \in R$ for all $a \in A$
2. Symmetric: $(a, b) \in R \rightarrow (b, a) \in R$

Note: Transitivity is NOT required. This is the key difference between an equivalence relation and a compatibility relation.

Example:

$A = \{1, 2, 3\}$ $R = \{(1,1), (2,2), (3,3), (1,2), (2,1), (1,3), (3,1)\}$

Reflexive: ✔ Symmetric: (1,2) and (2,1) ✔, (1,3) and (3,1) ✔ Transitive? $(2,1) \in R$ and $(1,3) \in R \rightarrow$ is $(2,3) \in R$? No ✗ → NOT transitive, but still a valid compatibility relation.

Real-Life Analogy: "Can communicate with" in a network — A can talk to B, B can talk to A, but A may not be able to talk to C even though B can talk to C. Compatibility but not transitivity.

10.2 Compatible Set

Definition: A subset S of A is called a compatible set with respect to a compatibility relation R if every pair of elements in S is related under R.

Formally: For all $a, b \in S$, $(a, b) \in R$

Example:

$A = \{1, 2, 3, 4\}$ $R = \{(1,1), (2,2), (3,3), (4,4), (1,2), (2,1), (1,3), (3,1), (2,4), (4,2)\}$

Is {1, 2} a compatible set? $(1,2) \in R$ ✔ and $(2,1) \in R$ ✔ → Yes, {1,2} is compatible.

Is {1, 2, 3} a compatible set? $(1,2) \in R$ ✔, $(1,3) \in R$ ✔, but $(2,3) \notin R$ ✗ → No, not compatible.

10.3 Maximum Compatibility Block (Maximal Compatible Set)

Definition: A compatible set S is called a maximum compatibility block (or maximal compatible set) if no additional element can be added to S while keeping it compatible.

That is, S is a maximal compatible set if there is no element $x \in A - S$ such that $\{S \cup \{x\}\}$ is still a compatible set.

Step-by-Step Method to Find Maximum Compatibility Blocks:

Step 1: Draw the compatibility graph — vertices are elements of A, edges connect pairs that are related under R (excluding self-loops).

Step 2: Find all maximal cliques in this graph (a clique is a set of vertices where every two vertices are connected by an edge).

Step 3: Each maximal clique corresponds to a maximum compatibility block.

Example:

A = {1, 2, 3, 4, 5} Compatible pairs (non-reflexive): {1,2}, {1,3}, {2,3}, {3,4}, {4,5}

Compatibility Graph edges: 1-2, 1-3, 2-3, 3-4, 4-5

Cliques: {1, 2, 3} — all three are pairwise related ✔ → maximal (cannot add 4 since $(1,4) \notin R$) {3, 4} — related ✔ → not maximal yet, can we add 5? $(3,5) \notin R$, so cannot. Can we add 1 or 2? No. → maximal {4, 5} — related ✔ → maximal

Maximum Compatibility Blocks: {1,2,3}, {3,4}, {4,5}

Key Point: Maximum compatibility blocks may overlap (an element can appear in more than one block). This is different from equivalence classes (which never overlap). The union of all maximum compatibility blocks must cover the entire set A.

11. Composite Relation

Definition: Let R be a relation from set A to set B, and S be a relation from set B to set C. The composite relation of R and S, denoted $R \circ S$ (or $S \circ R$ in some notations), is a relation from A to C defined as:

$R \circ S = \{(a, c) \mid \text{there exists } b \in B \text{ such that } (a, b) \in R \text{ and } (b, c) \in S\}$

In simple words: a is related to c in the composite if there is some "bridge" element b such that a → b under R and b → c under S.

Example:

A = {1, 2, 3}, B = {a, b, c}, C = {x, y}

R (from A to B) = {(1, a), (2, b), (3, c)} S (from B to C) = {(a, x), (b, x), (c, y)}

Find $R \circ S$:

$(1, a) \in R$ and $(a, x) \in S \rightarrow (1, x) \in R \circ S$ $(2, b) \in R$ and $(b, x) \in S \rightarrow (2, x) \in R \circ S$ $(3, c) \in R$ and $(c, y) \in S \rightarrow (3, y) \in R \circ S$

$R \circ S = \{(1,x), (2,x), (3,y)\}$

Matrix Method: The matrix of $R \circ S$ is obtained by Boolean matrix multiplication:

$M(R \circ S) = M(R) \otimes M(S)$

Boolean multiplication rule: Multiply row i of M(R) with column j of M(S) using AND, then combine results using OR.

Properties of Composite Relation:

1. Composition is generally NOT commutative: $R \circ S \neq S \circ R$ (in most cases)
2. Composition IS associative: $(R \circ S) \circ T = R \circ (S \circ T)$
3. If R is a relation on A, then $R^2 = R \circ R$ (relation composed with itself)

Real-Life Analogy: If "is a parent of" is R and "is a parent of" is S, then $R \circ S$ is "is a grandparent of."

12. Converse of a Relation

Definition: The converse of a relation R from set A to set B, denoted R^{-1} or R^T or $\tilde{R}$, is a relation from B to A obtained by reversing every ordered pair in R.

$R^{-1} = \{(b, a) \mid (a, b) \in R\}$

In simple words: swap the first and second elements of every pair.

Example:

$R = \{(1, a), (2, b), (3, a)\}$

$R^{-1} = \{(a, 1), (b, 2), (a, 3)\}$

Matrix of the Converse: The matrix of R^{-1} is the transpose of the matrix of R.

$M(R^{-1}) = M(R)^T$

Properties of Converse:

1. $(R^{-1})^{-1} = R$ (double converse gives back the original relation)
2. $(R \cup S)^{-1} = R^{-1} \cup S^{-1}$
3. $(R \cap S)^{-1} = R^{-1} \cap S^{-1}$
4. $(R \circ S)^{-1} = S^{-1} \circ R^{-1}$ (note: order reverses)
5. If R is symmetric, then $R^{-1} = R$

Real-Life Example: If R is "is a parent of", then R^{-1} is "is a child of." If R is "is greater than (>)", then R^{-1} is "is less than (<)."

13. Transitive Closure of a Relation

13.1 Definition

The transitive closure of a relation R on set A, denoted R^+ or t(R), is the smallest relation on A that:

1. Contains R (i.e., $R \subseteq R^+$)
2. Is transitive

In simple words: the transitive closure is obtained by adding all the "missing" pairs that are needed to make the relation transitive, while adding as few pairs as possible.

Intuition: If you can reach element c from element a by following a chain of related pairs (a→b→c, or a→b→c→d, etc.), then (a, c) must be in the transitive closure.

13.2 How to Compute Transitive Closure

Method 1 — Step-by-Step Composition:

Compute: $R^1 = R$, $R^2 = R \circ R$, $R^3 = R \circ R^2$, ... until no new pairs are added.

Transitive Closure: $R^+ = R^1 \cup R^2 \cup R^3 \cup \ldots$

For a set with n elements, at most n steps are needed.

Example:

A = {1, 2, 3} R = {(1,2), (2,3)}

R^1 = {(1,2), (2,3)} $R^2 = R \circ R$: (1,2) ∈ R and (2,3) ∈ R → (1,3) ∈ R^2 R^2 = {(1,3)}

$R^3 = R^2 \circ R$: (1,3) ∈ R^2 but (3,?) ∉ R → nothing new $R^3 = \varnothing$

$R^+ = R^1 \cup R^2$ = {(1,2), (2,3), (1,3)} ✔

Now check: is {(1,2), (2,3), (1,3)} transitive? (1,2) and (2,3) → (1,3) ∈ R^+ ✔ → Yes, transitive.

Method 2 — Warshall's Algorithm (Matrix Method):

This is the most efficient and commonly asked exam method.

Given: Adjacency matrix M of relation R on set A = $\{v_1, v_2, \ldots, v_n\}$

Algorithm:

Initialize: W_0 = M (start with the adjacency matrix)

For k = 1 to n: For i = 1 to n: For j = 1 to n: W_k[i][j] = W_{k-1}[i][j] OR (W_{k-1}[i][k] AND W_{k-1}[k][j])

Final result: W_n = Transitive Closure Matrix (Reachability Matrix)

Meaning of the formula: At step k, we check: can we reach j from i by going through vertex k as an intermediate? If yes, set W[i][j] = 1.

Example — Warshall's Algorithm:

A = {1, 2, 3} R = {(1,2), (2,3)}

Step 0 — Initial Matrix W_0:

	1	**2**	**3**
1	0	1	0
2	0	0	1
3	0	0	0

Step 1 — k=1 (using vertex 1 as intermediate):

For each (i,j): $W_1[i][j] = W_0[i][j]$ OR ($W_0[i][1]$ AND $W_0[1][j]$)

Check: Can any path go through vertex 1? $W_0[2][1] = 0$ → no path from 2 reaches 1, so no new entries added. $W_1 = W_0$ (no change)

Step 2 — k=2 (using vertex 2 as intermediate):

$W_1[i][j] = W_1[i][j]$ OR ($W_1[i][2]$ AND $W_1[2][j]$)

Check (i=1, j=3): W[1][2]=1 AND W[2][3]=1 → W[1][3] = 0 OR 1 = 1 ← NEW ENTRY

W_2:

	1	**2**	**3**
1	0	1	1
2	0	0	1
3	0	0	0

Step 3 — k=3 (using vertex 3 as intermediate):

$W_2[i][3]=0$ for all i where $W_2[3][j]$ might help → no new entries.

Final Reachability Matrix $W_3 = W_2$:

	1	**2**	**3**
1	0	1	1
2	0	0	1
3	0	0	0

This means: 1 can reach 2 and 3 ✔ 2 can reach 3 ✔ 3 cannot reach anyone ✔

Transitive Closure $R^+ = \{(1,2), (1,3), (2,3)\}$ ✔ (matches our manual result)

13.3 Reflexive Transitive Closure

If we also want the closure to be reflexive (include all self-pairs), we compute:

$R^* = I \cup R^+$

Where I is the identity relation $\{(a,a) \mid a \in A\}$.

In Warshall's matrix: set all diagonal entries to 1 in the starting matrix W_0, then run the algorithm.

Summary — All Key Concepts in Unit 4 Part A

Concept	Key Idea	Exam Focus
Relation	Subset of A × B	Definition and examples
Binary Relation	Relation on A × A	Properties check
Domain & Range	First and second elements of pairs	Finding from given R
Universal Relation	R = A × A	All n^2 pairs
Void Relation	R = ∅	No pairs
Union R ∪ S	Pair in R or S	OR of matrices
Intersection R ∩ S	Pair in both R and S	AND of matrices

Complement R'	Pair not in R	NOT of matrix
Reflexive	(a,a) ∈ R for all a	Diagonal all 1s
Symmetric	(a,b) → (b,a)	$M = M^T$
Anti-Symmetric	(a,b),(b,a) → a=b	No symmetric off-diagonal 1s
Transitive	(a,b),(b,c) → (a,c)	Chain check
Partition	Disjoint cover of A	Blocks
Equivalence Relation	Reflexive + Symmetric + Transitive	Equivalence classes = partition
Compatibility Relation	Reflexive + Symmetric only	Max compatibility blocks
Composite Relation	Bridge element method	Boolean matrix multiplication
Converse Relation	Reverse all pairs	Transpose of matrix
Transitive Closure	Smallest transitive superset	Warshall's Algorithm

Here is the complete, exam-ready text content for Unit 4 — Part B. Copy-paste directly into your Word file.

Unit 4 — Part B: Partial Ordering and Lattices

14. Partial Ordering (POSET)

14.1 Definition

A relation R on a set A is called a partial order relation if it satisfies all three of the following properties simultaneously:

1. Reflexive: (a, a) ∈ R for all a ∈ A
2. Anti-Symmetric: if (a, b) ∈ R and (b, a) ∈ R, then a = b
3. Transitive: if (a, b) ∈ R and (b, c) ∈ R, then (a, c) ∈ R

A set A together with a partial order relation R defined on it is called a Partially Ordered Set, or simply a POSET.

Notation: A POSET is denoted as (A, R) or more commonly (A, ≤) where ≤ represents the partial order relation (not necessarily the "less than or equal to" of numbers — it is a general symbol for any partial order).

The symbol a ≤ b is read as "a is related to b" or "a precedes b" or "a is less than or equal to b" in the context of the given partial order.

Why "Partial"? The word "partial" means that not every pair of elements needs to be comparable. It is perfectly valid in a POSET for two elements a and b to have no ordering between them — that is, neither $a \le b$ nor $b \le a$.

14.2 Examples of Partial Order Relations

Example 1 — Divisibility on Positive Integers:

Let $A = \{1, 2, 3, 4, 6, 12\}$ R = "a divides b" (written as $a \mid b$)

Check the three properties: Reflexive: Every number divides itself → $a \mid a$ ✔ Anti-Symmetric: If $a \mid b$ and $b \mid a$, then $a = b$ (for positive integers) ✔ Transitive: If $a \mid b$ and $b \mid c$, then $a \mid c$ ✔

So $(A, \mid)$ is a POSET.

Comparable pairs in this POSET: $1 \mid 2$, $1 \mid 3$, $1 \mid 4$, $1 \mid 6$, $1 \mid 12$ $2 \mid 4$, $2 \mid 6$, $2 \mid 12$ $3 \mid 6$, $3 \mid 12$ $4 \mid 12$, $6 \mid 12$

Non-comparable pairs (neither divides the other): $\{2, 3\}$, $\{3, 4\}$, $\{4, 6\}$ — these pairs have no ordering between them.

Example 2 — Subset Relation on Power Set:

Let $A = \{a, b\}$ and $P(A) = \{\varnothing, \{a\}, \{b\}, \{a,b\}\}$ R = "is a subset of" ($\subseteq$)

Reflexive: Every set is a subset of itself ✔ Anti-Symmetric: If $X \subseteq Y$ and $Y \subseteq X$, then $X = Y$ ✔ Transitive: If $X \subseteq Y$ and $Y \subseteq Z$, then $X \subseteq Z$ ✔

So $(P(A), \subseteq)$ is a POSET.

Comparable pairs: $\varnothing \subseteq \{a\}$, $\varnothing \subseteq \{b\}$, $\varnothing \subseteq \{a,b\}$, $\{a\} \subseteq \{a,b\}$, $\{b\} \subseteq \{a,b\}$ Non-comparable pairs: $\{a\}$ and $\{b\}$ (neither is a subset of the other)

Example 3 — Less Than or Equal to on Integers:

Let $A = \{1, 2, 3, 4\}$ R = "$\le$" (less than or equal to)

Reflexive: $a \le a$ ✔ Anti-Symmetric: if $a \le b$ and $b \le a$, then $a = b$ ✔ Transitive: if $a \le b$ and $b \le c$, then $a \le c$ ✔

So $(A, \le)$ is a POSET. In fact, in this case every pair is comparable — this leads us to the concept of a total order.

14.3 Comparable and Incomparable Elements

In a POSET $(A, \le)$:

Two elements a and b are called **comparable** if either $a \le b$ or $b \le a$ (at least one holds).

Two elements a and b are called **incomparable** if neither $a \le b$ nor $b \le a$ holds. This is written as $a \parallel b$.

Example:

In the POSET ({1,2,3,4,6,12}, |) — divisibility: 2 and 3 are incomparable because 2 does not divide 3 and 3 does not divide 2. 2 and 4 are comparable because $2 \mid 4$.

15. Simple / Linear Ordering and Totally Ordered Set (Chain)

15.1 Definition of Total Order (Linear Order)

A partial order relation ≤ on a set A is called a total order or linear order if every pair of elements in A is comparable.

Formally: For all $a, b \in A$, either $a \le b$ or $b \le a$ (or both, which means $a = b$).

A set A with a total order is called a totally ordered set or a chain.

Key Difference — Partial Order vs Total Order:

Property	Partial Order (POSET)	Total Order (Chain)
Reflexive	Required ✔	Required ✔
Anti-Symmetric	Required ✔	Required ✔
Transitive	Required ✔	Required ✔
Every pair comparable	NOT required	Required ✔

A total order is a stricter version of a partial order. Every total order is a partial order, but not every partial order is a total order.

15.2 Examples of Total Order

Example 1: A = {1, 2, 3, 4} with R = "≤"

$1 \le 2 \le 3 \le 4$ — every pair is comparable → This is a total order (chain).

Example 2: A = {1, 2, 3, 6} with R = "a | b" (divisibility)

Is 2 comparable with 3? 2 does not divide 3, and 3 does not divide 2 → $2 \parallel 3$ → NOT a total order. It is only a partial order.

Example 3 — A Chain within a POSET: In the POSET ({1,2,4,8}, |): $1 \mid 2 \mid 4 \mid 8$ — every element divides the next → This is a chain (totally ordered subset).

15.3 Frequently Used Partial Order Relations

The following are the most commonly encountered partial order relations in mathematics and computer science:

Relation	Set	Symbol	Description
Less than or equal to	Integers, Reals	≤	a ≤ b means a is at most b

Divisibility	Positive Integers	|	a | b means a divides b
Subset	Power set of any set	⊆	A ⊆ B means A is contained in B
Alphabetical / Lexicographic order	Strings, Words	≤_lex	Dictionary ordering
Prefix ordering	Strings	prefix	String a is a prefix of string b

All of these are partial orders. Among them, "≤" on integers is also a total order.

16. Representation of Partially Ordered Sets — Hasse Diagram

16.1 Why Hasse Diagrams?

A POSET can be represented as a directed graph, but such a graph quickly becomes cluttered with many edges (especially reflexive loops and transitive edges that are "implied"). The Hasse diagram is a simplified, clean visual representation of a POSET that removes all unnecessary edges while preserving all the ordering information.

16.2 Rules for Drawing a Hasse Diagram

To draw the Hasse diagram of a POSET (A, ≤):

Step 1 — Remove all reflexive edges (self-loops). Since a ≤ a for all a, these are implied and do not need to be drawn.

Step 2 — Remove all transitive edges. If a ≤ b and b ≤ c, then we already know a ≤ c. So the direct edge a → c is redundant and is removed.

Step 3 — Remove arrow directions. Instead of arrows, use the position of elements to indicate direction. If a ≤ b, then b is drawn above a. Going upward in the diagram means "greater than."

Step 4 — Draw edges only between directly related elements. An edge is drawn between a and b (with b above a) only if a ≤ b and there is no element c such that a ≤ c ≤ b (with c different from a and b). Such a pair is called a covering relation: b covers a.

Summary — What Hasse Diagram Removes vs Keeps:

Type of Edge	In Full Digraph	In Hasse Diagram
Self-loops (reflexive)	Present	Removed
Transitive edges	Present	Removed
Direct covering edges	Present	Kept
Arrow heads	Present	Removed (use position)

16.3 Example — Drawing a Hasse Diagram

Example 1:

POSET: A = {1, 2, 3, 4, 6, 12}, R = divisibility (|)

First identify all covering relations (b covers a means a | b and no element sits strictly between them):

1 is covered by 2, 3 2 is covered by 4, 6 3 is covered by 6 4 is covered by 12 6 is covered by 12

Hasse Diagram (drawn from bottom to top, lowest element at bottom):

```
  12
 / \
4   6
|   /\
2   3   (2 already placed)
 \ |
  1
```

Reading the diagram: Going upward from 1: 1 divides 2, 3 (directly above 1) Going upward from 2: 2 divides 4 and 6 Going upward from 3: 3 divides 6 Going upward from 4 and 6: both divide 12

Example 2:

POSET: P({a,b,c}) — power set of {a,b,c} with $\subseteq$

Elements: $\emptyset$, {a}, {b}, {c}, {a,b}, {a,c}, {b,c}, {a,b,c}

Covering relations: $\emptyset$ is covered by {a}, {b}, {c} {a} is covered by {a,b}, {a,c} {b} is covered by {a,b}, {b,c} {c} is covered by {a,c}, {b,c} {a,b}, {a,c}, {b,c} are all covered by {a,b,c}

Hasse Diagram (levels from bottom to top):

Level 0 (bottom): $\emptyset$ Level 1: {a}, {b}, {c} Level 2: {a,b}, {a,c}, {b,c} Level 3 (top): {a,b,c}

16.4 Reading a Hasse Diagram

Given a Hasse diagram, you can recover the full relation R by the following rule: a $\leq$ b if and only if there is a path going upward from a to b in the Hasse diagram.

Example: From the diagram of ({1,2,3,4,6,12}, |): 1 $\leq$ 12 because there is an upward path 1 → 2 → 4 → 12 (or 1 → 2 → 6 → 12, etc.) 3 $\leq$ 12 because there is an upward path 3 → 6 → 12 4 and 6 are incomparable because there is no upward path from 4 to 6 or from 6 to 4.

17. Least, Greatest, Minimal, and Maximal Elements

These are four distinct concepts. Students often confuse minimal with least, and maximal with greatest. Understanding the difference is critical for exams.

17.1 Least Element

Definition: An element a ∈ A is called the least element (also called the minimum element or bottom element) of the POSET (A, ≤) if a ≤ x for every x ∈ A.

In other words: a is related to (precedes) every other element in the set. It is the single smallest element.

Key Points: A POSET can have at most one least element (if it exists). The least element is comparable to every element in A. The least element is always a minimal element too.

Example:

POSET ({1,2,3,4,6,12}, |): 1 divides every other element → 1 is the least element.

POSET (P({a,b}), ⊆): ∅ ⊆ every set → ∅ is the least element.

17.2 Greatest Element

Definition: An element b ∈ A is called the greatest element (also called the maximum element or top element) of the POSET (A, ≤) if x ≤ b for every x ∈ A.

In other words: every element in A precedes b. It is the single largest element.

Key Points: A POSET can have at most one greatest element (if it exists). The greatest element is comparable to every element in A.

Example:

POSET ({1,2,3,4,6,12}, |): Every element divides 12 → 12 is the greatest element.

POSET (P({a,b}), ⊆): Every set is a subset of {a,b} → {a,b} is the greatest element.

17.3 Minimal Element

Definition: An element a ∈ A is called a minimal element of the POSET (A, ≤) if there is no element x ∈ A such that x ≤ a and x ≠ a.

In other words: nothing in A comes strictly before a. There is no element smaller than a.

Key Points: A POSET can have multiple minimal elements. A minimal element is NOT required to be comparable with every other element. If a least element exists, it is the only minimal element.

Example 1 — Single Minimal Element:

POSET ({1,2,3,4,6,12}, |): Nothing divides 1 (except 1 itself) → 1 is the only minimal element.

Example 2 — Multiple Minimal Elements:

A = {2, 3, 4, 6} with divisibility 2 divides 4 and 6. 3 divides 6. Nothing in A divides 2 (within A). Nothing in A divides 3 (within A). → Both 2 and 3 are minimal elements.

Note: 2 and 3 are incomparable with each other, which is why both can be minimal.

17.4 Maximal Element

Definition: An element b ∈ A is called a maximal element of the POSET (A, ≤) if there is no element x ∈ A such that b ≤ x and x ≠ b.

In other words: nothing in A comes strictly after b. There is no element larger than b.

Key Points: A POSET can have multiple maximal elements. A maximal element is NOT required to be comparable with every other element. If a greatest element exists, it is the only maximal element.

Example:

A = {2, 3, 4, 6} with divisibility: 4 is not divided by anything in A except itself → 4 is maximal. 6 is not divided by anything in A except itself → 6 is maximal. → Both 4 and 6 are maximal elements.

17.5 Key Comparison Table

Feature	Least Element	Minimal Element	Greatest Element	Maximal Element
Definition	a ≤ x for ALL x	No x < a exists	x ≤ b for ALL x	No x > b exists
Uniqueness	At most one	Can be many	At most one	Can be many
Comparability	Comparable to all	May be incomparable to some	Comparable to all	May be incomparable to some
Relation to other	Least ⊆ Minimal	Minimal ⊇ Least	Greatest ⊆ Maximal	Maximal ⊇ Greatest

Memory Trick: Least / Greatest → Global champions (they beat everyone or lose to everyone). Minimal / Maximal → Local champions (no one beats them in their neighbourhood, but they may not be comparable to all).

18. Least Upper Bound (Supremum) and Greatest Lower Bound (Infimum)

These concepts apply when we look at a subset B of a POSET (A, ≤) and ask: what is the smallest element that is "above" everything in B, and what is the largest element "below" everything in B?

18.1 Upper Bound

Definition: Let (A, ≤) be a POSET and B ⊆ A. An element u ∈ A is called an upper bound of B if b ≤ u for every b ∈ B.

In other words: u is "above" (or equal to) every element of B.

Example:

POSET ({1,2,3,4,6,12}, |), B = {2, 3}

Which elements of A are divisible by both 2 and 3? 6 is divisible by 2 and 3 ✔ → 6 is an upper bound. 12 is divisible by 2 and 3 ✔ → 12 is an upper bound.

Upper bounds of B = {2,3} are: {6, 12}

18.2 Least Upper Bound (LUB / Supremum / Join)

Definition: The least upper bound (LUB) of a subset B, denoted sup(B) or ∨B (join), is the smallest element among all upper bounds of B.

That is: u* is the LUB of B if: Condition 1: u* is an upper bound of B (b ≤ u* for all b ∈ B) Condition 2: u* ≤ u for every other upper bound u of B

Key Points: The LUB may or may not exist in a given POSET. If the LUB exists, it is unique. For two elements a and b, LUB({a,b}) is written as a ∨ b (read "a join b").

Example:

POSET ({1,2,3,4,6,12}, |), B = {2, 3}

Upper bounds of {2,3}: 6 and 12. Among 6 and 12: 6 | 12 → 6 is smaller. LUB({2,3}) = 6 ✔

(6 is the smallest element that both 2 and 3 divide into.)

18.3 Lower Bound

Definition: An element l ∈ A is called a lower bound of B ⊆ A if l ≤ b for every b ∈ B.

In other words: l is "below" (or equal to) every element of B.

Example:

POSET ({1,2,3,4,6,12}, |), B = {4, 6}

Which elements divide both 4 and 6? 1 divides both ✔ → 1 is a lower bound. 2 divides both ✔ → 2 is a lower bound.

Lower bounds of {4,6}: {1, 2}

18.4 Greatest Lower Bound (GLB / Infimum / Meet)

Definition: The greatest lower bound (GLB) of a subset B, denoted inf(B) or ∧B (meet), is the largest element among all lower bounds of B.

That is: l* is the GLB of B if: Condition 1: l* is a lower bound of B (l* ≤ b for all b ∈ B) Condition 2: l ≤ l* for every other lower bound l of B

Key Points: The GLB may or may not exist in a given POSET. If the GLB exists, it is unique. For two elements a and b, GLB({a,b}) is written as a ∧ b (read "a meet b").

Example:

POSET ({1,2,3,4,6,12}, |), B = {4, 6}

Lower bounds of {4,6}: 1 and 2. Among 1 and 2: 1 | 2 → 2 is larger. GLB({4,6}) = 2 ✔

(2 is the greatest element that divides both 4 and 6 — this is the GCD.)

18.5 Real-Life Meaning of LUB and GLB

In divisibility POSET: LUB of {a, b} = LCM (Least Common Multiple) of a and b GLB of {a, b} = GCD (Greatest Common Divisor) of a and b

In subset POSET (P(A), $\subseteq$): LUB of {X, Y} = $X \cup Y$ (union) GLB of {X, Y} = $X \cap Y$ (intersection)

18.6 Summary Table — Bounds

Concept	Symbol	Meaning	Also Called
Upper Bound of B	u	$b \le u$ for all $b \in B$	—
Least Upper Bound	sup(B) or $a \vee b$	Smallest upper bound	Join, Supremum
Lower Bound of B	l	$l \le b$ for all $b \in B$	—
Greatest Lower Bound	inf(B) or $a \wedge b$	Largest lower bound	Meet, Infimum

19. Well-Ordered POSET

Definition: A POSET (A, ≤) is called well-ordered if it is a totally ordered set (chain) AND every non-empty subset of A has a least element.

Two conditions required:

1. Total Order: Every pair of elements is comparable.
2. Every Non-Empty Subset Has a Least Element: There is always a minimum in every subset.

Example 1 — Well-Ordered:

($\mathbb{N}$, ≤) — the set of natural numbers {0, 1, 2, 3, ...} with the usual ≤ ordering. Every non-empty subset of $\mathbb{N}$ has a smallest element. ✔ → Well-ordered.

Example 2 — Not Well-Ordered:

($\mathbb{Z}$, ≤) — the set of all integers with ≤. The subset of all negative integers {-1, -2, -3, ...} has no smallest element. ✗ → NOT well-ordered.

Example 3 — Not Well-Ordered:

($\mathbb{R}$, ≤) — real numbers with ≤. The open interval (0, 1) has no least element. ✗ → NOT well-ordered.

Key Property: Every finite totally ordered set is automatically well-ordered.

Real-Life Importance: Well-ordering is the foundation of mathematical induction and is used in theoretical computer science for proving program termination.

20. Lattice as a POSET

20.1 Definition

A POSET (A, ≤) is called a lattice if every pair of elements a, b ∈ A has:

1. A unique Least Upper Bound (LUB): a ∨ b exists in A
2. A unique Greatest Lower Bound (GLB): a ∧ b exists in A

In simple words: a lattice is a POSET where any two elements can always be "joined" (LUB) and "met" (GLB), and both results stay within the set.

The operations ∨ (join) and ∧ (meet) are the two fundamental operations of a lattice.

Note: The condition is for every pair of elements — not just some pairs.

20.2 Example — Checking if a POSET is a Lattice

Example 1 — Is a Lattice:

POSET ({1,2,3,6}, |) — divisibility.

Check all pairs for LUB and GLB:

Pair	LUB (∨)	GLB (∧)
{1,2}	2	1
{1,3}	3	1
{2,3}	6	1
{2,6}	6	2
{3,6}	6	3
{1,6}	6	1

Every pair has both a LUB and GLB within the set. ✔ → ({1,2,3,6}, |) is a lattice.

Example 2 — Not a Lattice:

A = {1, 2, 3, 4, 5} with divisibility.

Check pair {2,3}: Upper bounds of {2,3} in A: need something divisible by both 2 and 3. LCM(2,3) = 6, but 6 ∉ A. → No LUB exists in A for the pair {2,3}. ✗ → NOT a lattice.

20.3 Lattice as an Algebraic System

A lattice can also be defined purely algebraically (without reference to a POSET), as a set A with two binary operations ∨ (join) and ∧ (meet) satisfying:

For all a, b, c ∈ A:

L1 — Commutative Laws: $a \vee b = b \vee a$ $a \wedge b = b \wedge a$

L2 — Associative Laws: $(a \vee b) \vee c = a \vee (b \vee c)$ $(a \wedge b) \wedge c = a \wedge (b \wedge c)$

L3 — Absorption Laws: $a \vee (a \wedge b) = a$ $a \wedge (a \vee b) = a$

L4 — Idempotent Laws: $a \vee a = a$ $a \wedge a = a$

These four pairs of laws are the defining properties of a lattice.

20.4 Properties of Lattice Operations

In any lattice $(A, \vee, \wedge)$:

1. Idempotent: $a \vee a = a$ and $a \wedge a = a$
2. Commutative: $a \vee b = b \vee a$ and $a \wedge b = b \wedge a$
3. Associative: $(a \vee b) \vee c = a \vee (b \vee c)$ and similarly for $\wedge$
4. Absorption: $a \vee (a \wedge b) = a$ and $a \wedge (a \vee b) = a$
5. Consistency with order: $a \le b$ if and only if $a \vee b = b$, equivalently $a \wedge b = a$

21. Types of Lattices

21.1 Complete Lattice

Definition: A lattice $(A, \le)$ is called a complete lattice if every subset $B \subseteq A$ (not just pairs, but any subset — including the empty set and the entire set) has both a LUB and a GLB in A.

Key difference from a regular lattice: A regular lattice only guarantees LUB and GLB for pairs of elements. A complete lattice guarantees them for all subsets.

Example 1 — Complete Lattice:

$(P(S), \subseteq)$ — the power set of any set S ordered by inclusion.

For any collection of subsets, their union is the LUB and their intersection is the GLB, and both remain within P(S). → Always a complete lattice.

Example 2 — Complete Lattice:

$([0, 1], \le)$ — the closed unit interval of real numbers with $\le$. LUB of any subset = supremum of the subset (maximum value). GLB of any subset = infimum of the subset (minimum value). Both exist in [0,1] for any subset. ✔ → Complete lattice.

Example 3 — Complete Lattice:

Any finite lattice is always a complete lattice because all subsets are finite and LUB/GLB always exist.

Key Property: In a complete lattice: LUB of the entire set A = greatest element (top element, denoted 1 or $\top$) GLB of the entire set A = least element (bottom element, denoted 0 or $\bot$) So every complete lattice has both a top and a bottom element.

Theorem: Every finite lattice is a complete lattice.

21.2 Distributive Lattice

Definition: A lattice $(A, \vee, \wedge)$ is called a distributive lattice if the join ($\vee$) distributes over meet ($\wedge$) and the meet ($\wedge$) distributes over join ($\vee$).

The two distributive laws are:

Law 1: $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$ Law 2: $a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$

Note: In any lattice, if Law 1 holds, then Law 2 also holds automatically, and vice versa. So you only need to verify one of them.

Example 1 — Distributive Lattice:

$(P(S), \subseteq)$ with $\vee = \cup$ and $\wedge = \cap$:

$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ ✔ — This is a standard set theory law. → Power set lattice is always distributive.

Example 2 — Distributive Lattice:

$(\{1,2,3,6\}, |)$ with $\vee$ = LCM and $\wedge$ = GCD:

Check: $2 \wedge (3 \vee 6)$ = GCD(2, LCM(3,6)) = GCD(2,6) = 2 $(2 \wedge 3) \vee (2 \wedge 6)$ = LCM(GCD(2,3), GCD(2,6)) = LCM(1,2) = 2 ✔ → Distributive.

Example 3 — NOT Distributive:

Consider the diamond lattice M_3 with elements $\{0, a, b, c, 1\}$ where: 0 is the bottom (least element) 1 is the top (greatest element) a, b, c are all incomparable to each other $a \vee b = a \vee c = b \vee c = 1$ $a \wedge b = a \wedge c = b \wedge c = 0$

Check: $a \wedge (b \vee c) = a \wedge 1 = a$ $(a \wedge b) \vee (a \wedge c) = 0 \vee 0 = 0$

$a \neq 0$ ✗ → M_3 is NOT distributive.

Theorem (Birkhoff's Theorem): A lattice is distributive if and only if it contains no sublattice isomorphic to the diamond lattice M_3 or the pentagon lattice N_5.

21.3 Modular Lattice

Definition: A lattice $(A, \vee, \wedge)$ is called a modular lattice if for all $a, b, c \in A$:

If $a \leq c$, then $a \vee (b \wedge c) = (a \vee b) \wedge c$

This is called the modular law. It is a weakened version of the distributive law — it only requires the distributive property to hold when $a \leq c$.

Relationship: Every distributive lattice is modular. But not every modular lattice is distributive.

Distributive $\subset$ Modular $\subset$ Lattice (hierarchy from strictest to least strict)

Example 1 — Modular but Not Distributive:

The diamond lattice M_3 ($\{0, a, b, c, 1\}$) is modular but not distributive.

Example 2 — Not Modular:

The pentagon lattice N_5 with elements {0, a, b, c, 1} where: $0 < a < c < 1$ and $0 < b < 1$, with b incomparable to a and c, but $a < c$ and $b < c$.

For $a \le c$: $a \vee (b \wedge c) = a \vee 0 = a$ (since b and c are ordered as $b < c$ so $b \wedge c = b$ — wait, checking specific values shows the law fails) $(a \vee b) \wedge c = c \wedge c = c$

$a \neq c$ ✗ → N_5 is NOT modular.

Theorem: A lattice is modular if and only if it contains no sublattice isomorphic to the pentagon lattice N_5.

Summary of Hierarchy:

Every distributive lattice is modular. Every modular lattice is a lattice. But the reverse is not true in general.

Distributive → Modular → Lattice

21.4 Complemented Lattice

Definition: A lattice $(A, \vee, \wedge)$ with a least element 0 and a greatest element 1 is called a complemented lattice if for every element $a \in A$, there exists at least one element $a' \in A$ (called the complement of a) such that:

Condition 1: $a \vee a' = 1$ (join with complement gives the top) Condition 2: $a \wedge a' = 0$ (meet with complement gives the bottom)

Key Points: A complemented lattice must have a least element (0) and a greatest element (1) — i.e., it must be bounded. The complement of an element may not be unique in a general complemented lattice. In a distributive complemented lattice, the complement is always unique.

Example 1 — Complemented Lattice:

$(P(S), \subseteq)$ with $\vee = \cup$, $\wedge = \cap$, $0 = \emptyset$, $1 = S$:

For any set $A \subseteq S$, its complement is $A' = S - A$. $A \cup A' = S = 1$ ✔ $A \cap A' = \emptyset = 0$ ✔ → Complemented lattice.

Example 2 — Complemented Lattice:

$(\{1,2,3,6\}, |)$ with 0 = 1 (the number 1, least element) and 1 = 6 (greatest element):

For element 2: its complement would need x such that LCM(2,x) = 6 and GCD(2,x) = 1. x = 3: LCM(2,3) = 6 ✔ and GCD(2,3) = 1 ✔ → 3 is the complement of 2. For element 3: complement is 2 (by same logic). ✔ → Complemented lattice.

Example 3 — NOT Complemented:

$(\{1,2,4,8\}, |)$: Least = 1, Greatest = 8. For element 4: need x such that LCM(4,x) = 8 and GCD(4,x) = 1. LCM(4,x) = 8 → x must divide 8 and be a multiple of 2 → $x \in \{2,4,8\}$. GCD(4,2) = 2 ≠ 1 ✗, GCD(4,4) = 4 ≠ 1 ✗, GCD(4,8) = 4 ≠ 1 ✗ → No complement for 4 exists. NOT complemented.

22. Boolean Lattice and Pseudo-Boolean Lattice

22.1 Boolean Lattice (Boolean Algebra as a Lattice)

Definition: A lattice $(A, \vee, \wedge)$ is called a Boolean lattice (or Boolean algebra) if it satisfies all of the following conditions:

1. It is a distributive lattice.
2. It is a complemented lattice (has 0, 1, and every element has a unique complement).

In other words:

Boolean Lattice = Distributive + Complemented

Since it is distributive and complemented, in a Boolean lattice: Every element has a unique complement. De Morgan's laws hold: $(a \vee b)' = a' \wedge b'$ and $(a \wedge b)' = a' \vee b'$

Laws of Boolean Lattice / Boolean Algebra:

For all a, b, c in a Boolean lattice:

Law	Join Form	Meet Form
Identity	$a \vee 0 = a$	$a \wedge 1 = a$
Domination	$a \vee 1 = 1$	$a \wedge 0 = 0$
Idempotent	$a \vee a = a$	$a \wedge a = a$
Complement	$a \vee a' = 1$	$a \wedge a' = 0$
Commutative	$a \vee b = b \vee a$	$a \wedge b = b \wedge a$
Associative	$(a \vee b) \vee c = a \vee (b \vee c)$	$(a \wedge b) \wedge c = a \wedge (b \wedge c)$
Distributive	$a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$	$a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$
De Morgan's	$(a \vee b)' = a' \wedge b'$	$(a \wedge b)' = a' \vee b'$
Double Complement	$(a')' = a$	—
Absorption	$a \vee (a \wedge b) = a$	$a \wedge (a \vee b) = a$

Example 1 — Boolean Lattice:

$(P(S), \subseteq)$ — power set lattice: Distributive: ✔ (union distributes over intersection) Complemented: ✔ (complement of set A is S − A) → Boolean lattice. ✔

Example 2 — Boolean Lattice (Simplest Non-Trivial):

$B_2 = (\{0, 1\}, \vee, \wedge)$ — the two-element Boolean algebra: $0 \vee 1 = 1$, $0 \wedge 1 = 0$ Complement of 0 is 1, complement of 1 is 0. Distributive ✔, Complemented ✔ → Boolean lattice. ✔

This is the basis of all binary logic in digital circuits.

Example 3 — Boolean Lattice:

The lattice of divisors of 30 = 2 × 3 × 5: Elements: {1, 2, 3, 5, 6, 10, 15, 30} with divisibility. This forms a Boolean lattice with 8 = 2^3 elements (corresponding to subsets of {2,3,5}).

Theorem — Stone's Representation Theorem: Every finite Boolean lattice is isomorphic to the power set lattice P(S) for some set S. If $|S| = n$, then the Boolean lattice has exactly 2^n elements.

Key Point: Every Boolean algebra must have 2^n elements for some non-negative integer n. A Boolean algebra with 1 element (trivial), 2 elements, 4 elements, 8 elements, 16 elements, etc. are all valid. A Boolean algebra with 3 elements, 5 elements, or 6 elements is NOT possible.

22.2 Pseudo-Boolean Lattice (Heyting Algebra)

Definition: A pseudo-Boolean lattice (also called a Heyting algebra) is a bounded distributive lattice $(A, \vee, \wedge, 0, 1)$ equipped with an additional binary operation called pseudo-complement or relative pseudo-complement, denoted $a \to b$, satisfying:

$x \le (a \to b)$ if and only if $x \wedge a \le b$

The pseudo-complement of an element a (relative to 0) is defined as:

$a^* = a \to 0$ = the largest element x such that $x \wedge a = 0$

Key Differences — Boolean vs Pseudo-Boolean:

Feature	Boolean Lattice	Pseudo-Boolean Lattice
Distributive	✔ Required	✔ Required
Complemented	✔ Required (classical complement)	Pseudo-complement only (may not be classical)
Complement law	$a \vee a' = 1$ always	$a \vee a^* = 1$ may NOT hold
De Morgan's Laws	Hold fully	May not hold
Law of Excluded Middle	$a \vee a' = 1$ ✔	May fail
Double complement	$(a')' = a$ ✔	$(a^*)^* \ge a$ (not necessarily equal)

Intuition: In a Boolean lattice, every element has a true classical complement — a sharp "yes or no" complement. In a pseudo-Boolean lattice, we have a weaker form of complement — the largest element that has "no overlap" with a, but joining a with its pseudo-complement may not give the top element 1.

Example 1 — Boolean is also Pseudo-Boolean:

Every Boolean lattice is automatically a pseudo-Boolean lattice (Heyting algebra), because the classical complement satisfies the pseudo-complement condition.

Example 2 — Pseudo-Boolean but NOT Boolean:

The lattice of open sets of any topological space (ordered by inclusion) is a pseudo-Boolean lattice. But it is not necessarily Boolean, because the complement of an open set (a closed set) is not necessarily open, so classical complements may not exist within the structure.

Example 3 — Simple Example:

Consider A = {0, a, b, 1} with 0 < a < 1, 0 < b < 1, and a, b incomparable.

Pseudo-complement of a: largest x such that $x \wedge a = 0$. If $a \wedge b = 0$, then $a^* = b$ (pseudo-complement of a is b). But is $a \vee a^* = a \vee b = 1$? Not necessarily — if $a \vee b \neq 1$, then it is pseudo-Boolean but not Boolean.

Why Pseudo-Boolean Lattices Matter: Pseudo-Boolean lattices provide the algebraic foundation for intuitionistic logic, which is a constructive form of logic used in proof theory, type theory, and functional programming languages like Haskell and Coq.

Summary of All Lattice Types

Lattice Type	Conditions Required	Example
Lattice	Every pair has LUB and GLB	({1,2,3,6}, |)
Complete Lattice	Every subset has LUB and GLB	(P(S), ⊆)
Modular Lattice	Modular law holds ($a \le c \to a\vee(b\wedge c) = (a\vee b)\wedge c$)	Diamond lattice M_3
Distributive Lattice	Distributive laws hold	({1,2,3,6}, |)
Complemented Lattice	Every element has a complement	(P(S), ⊆)
Boolean Lattice	Distributive + Complemented	(P(S), ⊆), B_2
Pseudo-Boolean Lattice	Distributive + Pseudo-complement	Open sets of topology

Hierarchy (from most general to most specific):

Lattice ⊃ Complete Lattice ⊃ Modular Lattice ⊃ Distributive Lattice ⊃ Complemented Distributive Lattice = Boolean Lattice

Pseudo-Boolean Lattice sits between Distributive Lattice and Boolean Lattice.

Final Master Summary — Unit 4 Part B

Concept	Definition in One Line	Key Exam Point
Partial Order	Reflexive + Anti-Symmetric + Transitive	Check all three properties
POSET	Set + Partial Order Relation	Notation: (A, ≤)
Total Order / Chain	Every pair is comparable	Stricter than partial order
Hasse Diagram	Remove loops, transitive edges, arrows	b above a means a ≤ b
Least Element	Precedes every element	Unique if exists
Greatest Element	Follows every element	Unique if exists
Minimal Element	Nothing strictly below it	Can have many

Maximal Element	Nothing strictly above it	Can have many
LUB (Supremum)	Smallest upper bound	a ∨ b, Join
GLB (Infimum)	Largest lower bound	a ∧ b, Meet
Well-Ordered	Total order + every subset has a least element	$\mathbb{N}$ is well-ordered, $\mathbb{Z}$ is not
Lattice	Every pair has LUB and GLB	Two operations: ∨ and ∧
Complete Lattice	Every subset has LUB and GLB	All finite lattices are complete
Modular Lattice	Modular law	Contains no N_5 sublattice
Distributive Lattice	Distributive laws	Contains no M_3 or N_5 sublattice
Complemented Lattice	Every element has a complement	Must have 0 and 1
Boolean Lattice	Distributive + Complemented	2^n elements, De Morgan holds
Pseudo-Boolean	Distributive + Pseudo-complement	Basis of intuitionistic logic

VIDEO BASED LEARNING :

Relations (Part A)

#	Topic	Link
1	Introduction to Relations	https://www.youtube.com/watch?v=h5Lv5ZeNl0g
2	Reflexive Relation with examples	https://www.youtube.com/watch?v=-6oHeRpVG_o
3	Antisymmetric Relation with examples	https://www.youtube.com/watch?v=d6yFAmcL2bo
4	Transitive Relation with examples	https://www.youtube.com/watch?v=aqqVDVPftzE
5	Partial Order Relation and POSET	https://www.youtube.com/watch?v=qC2reWAo2mc

Full Gate Smashers DM Playlist (covers all relation topics sequentially):
https://www.youtube.com/playlist?list=PLxCzCOWd7aiH2wwES9vPWsEL6ipTaUSl3

https://youtube.com/playlist?list=PLEHGYFbPuuMF69CYt_IXz4CPOF44gNtSR&si=s0mSoCKOoDtP-GOz/

Hasse Diagram Specific Videos

#	Topic	Channel	Link
1	Hasse Diagram — Draw and Explain	Mathematics	https://www.youtube.com/watch?v=BHgu8a52jgM

	(Hindi)	Tutorial by Neha	
2	Hasse Diagram Problem 1 — POSET and Lattice	Ekeeda (Prof. Farhan Meer)	https://www.youtube.com/watch?v=18-mTB4pt-M
3	Hasse Diagram Problem 2 — POSET and Lattice	Ekeeda (Prof. Farhan Meer)	https://www.youtube.com/watch?v=kdqGGI2PBpo
4	Hasse Diagram and Lattices	GeeksForGeeks GATE	https://www.youtube.com/watch?v=r0Btow-bU_s

Dream Maths videos :

POSET: https://youtu.be/cbyWz_JYEhU?si=kB8b1JbM97vXCl0m/

Relation: https://youtu.be/y0TZi1ffFPY?si=nE4nfX5B-qygRRAZ/